Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023
Mid-Semester Test (EC-2 Makeup)

Course No. : AIMLCZG512 **Course Title** : Deep Reinforcement Learning
Nature of Exam : Open Book **Weightage** : 30% **No. of Pages** = 3 ; **No. Of Questions =** 4;
Duration : 2 Hours; **Date of Exam** : 21 December, 2025 - 09:00 AM

Note to Students:

1. Answer all the questions. All parts of a question should be answered consecutively. Each answer should start on a fresh page.
2. Write answers neatly in A4 paper, scan and upload.
3. Write your name and sign at the end of all the pages.
4. Assumptions made if any, should be stated clearly at the beginning of your answer.

Question -1 **[1.5 + 2+ 4 = 7.5 Marks]**

A mobile health-monitoring wristband must choose among three modes each hour, based on the current activity {resting, moderate, high}:

- Mode A: low power, moderate accuracy
- Mode B: medium power, high accuracy
- Mode C: high power, very high accuracy

The immediate benefit (reward) from each mode changes based on the user's physical activity.

a) State why RL (not supervised learning) fits this setting, and give a one-line distinction between immediate reward and long-term value. [1.5 Mark]

b) Using the *same wristband scenario*:

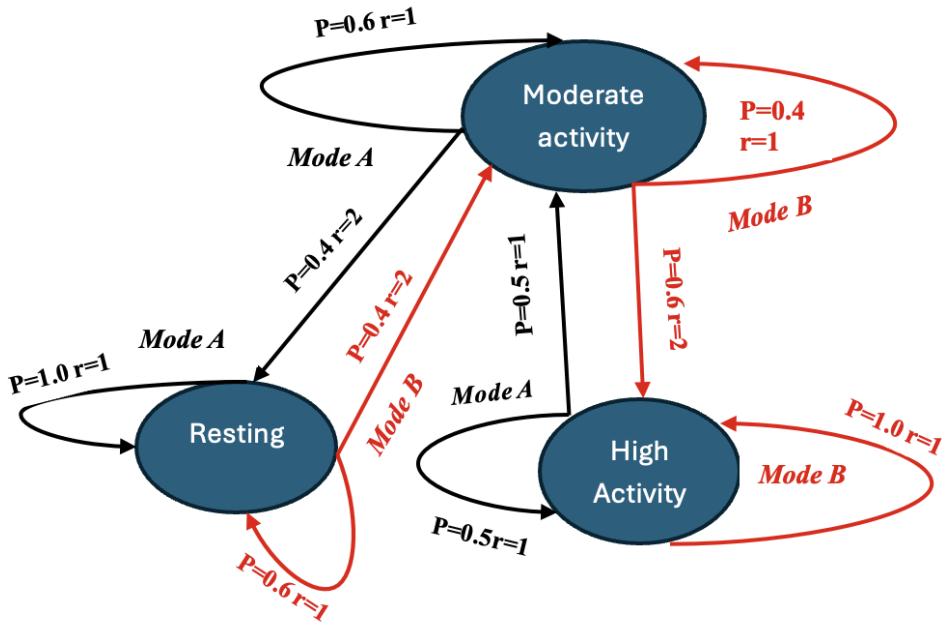
Case 1: Initially, assume the user's activity pattern is independent of the modes chosen, i.e., choosing Mode A or B today does not affect future rewards.

Case 2: Later, a firmware update introduces an adaptation rule: *If the wristband samples in Mode C for too long, the battery temperature rises, reducing future rewards for power-hungry modes.*

i) Classify case 1 as a Multi-Armed Bandit (MAB) or a Finite MDP, with one justification. [1 mark]

ii) Classify case 2 again and justify — does the added dependency change the problem class? [1 mark]

c) For the MDP given below, calculate the Values of states = {resting, moderate activity, high activity} in the same order, using synchronous Value Iteration. Actions = {mode A, mode B} are allowed from each state. Use $\gamma = 0.1$. Solve for 1 iteration. [4 marks]



Question -2 **[2.5+2+1.5+1.5 = 7.5 marks]**

In an ICU environment, clinicians must periodically adjust a patient's ventilator settings to maintain optimal blood oxygen saturation (SpO_2). The system can be modeled as a finite MDP where the patient's oxygenation levels and respiratory indicators are defined into categories such as lowO2, highO2, or OptimalO2. The clinician-assist AI can increase pressure, decrease pressure, or maintain settings. With 0.4 probability, increasing/decreasing the pressure does not change a patient's health. Whereas maintaining the same settings always seem to retain the patient's condition as is. The goal is to stabilize the patient's SpO_2 within safe clinical thresholds while avoiding over-ventilation. Once for four consecutive recordings, the SpO_2 is observed to be Optimal, then the patient stabilizes.

(a) Formulate the MDP. State the components clearly. Diagrammatically represent the model dynamics. **Note**: Assume equal likelihood for scenarios that are not explicitly mentioned in the above use case. [2.5 Marks]

(b) Given below reward designs, analyse the impact on the patient's safety in the given setup. Which one do you prefer and why? Justify your answer in exactly two statement w.r.t i)What behavior each encourage, ii) In which scenario one design is preferable over the other

Design A: For moderating from lowO2 to OptimalO2, reward = +10 is set

Design B: For moderating from OptimalO2 to HighO2, reward = -10 is set

[2 Marks]

(c) Is the given use case, episodic or continuous tasks? Justify your answer in no more than 30 words. [1.5 Marks]

(d) If $\gamma = 0.01$ is used, what is the impact on the patient's stability? Justify your answer in no more than 30 words. [1.5 Marks]

Question -3 **[3+1.5+1+2= 7.5 Marks]**

In an online pharma platform, a multi-armed bandit is implemented that recommends adherence-support interventions for (patients) frequent online customers based on their current state (e.g., health conditions, stock of medication, risk of the ordered medicines etc.,) to improve quality adherence and customer retention. Few observed interventions recommended for patients (customers) in a digital health platform include *{SMS reminder to refill medications(**S**), Teleconsultation Scheduling with a doctor (**T**), Reminder about Lab Tests (**R**), Counselling Call by Pharmacist (**C**)}*. Analyze the below observations in the given order and answer the following questions sequentially.

(1, S, 7), (2, T, 5), (3, R, 6), (4, C, 4), (5, S, 8), (6, T, 6), (7, R, 7), (8, C, 5).

a) Estimate the best intervention based on the listed observations. Use exponential recency weighted average based value update only with α=0.5, to revise the Q(actions) for all the actions. Assume for initialization. [3 Marks]

b) Refer to the results of part a) and illustrate the significance of α. What happens if α=1 is set instead? Justify answers in no more than 30 words using the given above use case for illustration. [1.5 Marks]

c) What is the significance of confidence level in the UCB based selection? Justify answers in no more than 30 words using the given above use case for illustrations. [1.0 Marks]

d) Analyze only until the 5th time step. On some of these time steps the ε-case may have occurred, causing an action to have been selected at random.

   i) On which time steps did this definitely occur? Why? [1.0Marks]

   ii) On which time steps could this possibly have occurred? Why? [1.0 Marks]

Question -4 **[2 + 2 + 3.5 = 7.5 Marks]**

a) In a model-free MDP, explain why estimating only the state value $v_\pi(s)$ is insufficient for selecting actions during control, even if v_π is estimated perfectly. [Ensure your answer is given in 2-3 well-articulated statements; vague answers will be penalised] [2 Marks]

b) In a deterministic policy that always selects the same action in a state, explain in 2-3 sentences why first-visit Monte Carlo estimation may fail to reach an optimal policy? Suggest one precise way to address this. [2 Marks]

c) A chatbot agent operates in two states: s_0 *for the user engaged* and s_1 *for the user disengaged*. It may take three actions: a_1 *ask an open-ended question*, a_2*: recommend a resource* and a_3*: send a fun fact*.

The present Q values and the ε-soft policy are as below:

$q_\pi(\ ,\ )$	a_1	a_2	a_3
s_0	1	0.5	0
s_1	0	0.5	1

$\pi(.\mid .)$	a_1	a_2	a_3
s_0	0.5	0.3	0.2
s_1	0.2	0.5	0.3

Now consider an episode:

$(s_0, a_1, r=2, s_0) \rightarrow (s_0, a_3, r=0, s_1) \rightarrow (s_1, a_2, r=3, s_1) \rightarrow (s_1, a_2, r=-1, \text{terminal})$.

Assuming γ=0.8 and computing returns using the discounted-sum formulation as per first visit Monte-Carlo method, produce the revised value-function table and the updated policy table at the end of this episode, clearly highlighting all changes. Show the computation corresponding to each change [3.5 Marks]